

Reading financial data

Les sottises des ignorans prennent autant de temps a lire, que les bonnes choses des doctes.
The nonsense of fools takes as much time to read as the good things of the learned.

From a letter by Constantijn Huygens to René Descartes.

There is now an enormous wealth of financial data available. Companies publish financial information in the form of accounts. The reporting period varies from country to country. In the US, such reports are required quarterly, while in the UK, companies are required to report twice each year and give details of their financial performance. The data within company reports is concerned with the fundamental health of a company and spans income from sales, expenditure on salaries, capital investment, overall profit or earning, dividend payouts to shareholders, and transfers to the company reserves. These reports together with periodic meetings with company directors are the greatest concern of *fundamental investors*, as opposed to *technical traders* who rely more on the asset price movements over time, rather than fundamental data. In this chapter we shall first describe the characteristics and general way data may be analysed. We then briefly describe these two different styles or approaches to investing.

2.1 Financial price data

Since the nineteenth century, financial transactions have been recorded on a daily basis, and since 1984, the sampling rate of transaction values has been made on a minute-by-minute basis. With the advent of modern computing technology, every transaction is now being recorded 'tick-by-tick', hence the term *tick data*. As a result, giga-amounts of data are now available. In his renowned 1963 study of cotton prices (Mandelbrot, 1963) that revealed for the first time a power law in the probability distribution function for price returns, the mathematician Benoît Mandelbrot used 2000 data points. In 1999, Gene Stanley and co-workers used 40 million points in a study of price fluctuations of 1000 companies quoted on the three major US stock markets: NSYE, AMEX, and NASDAQ (Gopikrishnan *et al.*, 1998).

Modern computers have not only allowed for more effective recording of prices but have also opened the way for more detailed analysis using tools of modern mathematics and physics. This is leading to a much greater understanding of market dynamics – as well as assisting applied studies of option pricing and portfolio management – as we discuss later in Chapters 8 and 11 respectively.

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2.1.1 Price scales, time scales, and trading time

In physics, reference units are maintained and improved by selected laboratories. This has led to the science of metrology. In finance, the quoted prices of stocks are generally in the form of currencies that fluctuate relative to other currencies. Moreover, currencies can change not only relative to other currencies, but also relative to assets such as gold. They are also affected by inflation, economic growth, or contraction (recession). The transactions (events) also occur at random times with random intensity. Physical time is well-defined. However, in finance, time scales take on an extra dimension. Although global markets are now active over the entire 24 hours of each day, there are still periods such as weekends and holidays where all the markets are closed and we really have no idea how to model such closed markets. For example, what is the effect of price sensitive news that arrives over such a closed period? An interesting variation on this is the continuing activity of an online betting market during half-time intervals of soccer matches (see Section 13.1). This is despite there being no news, as the players are off the pitch!

For currency transactions, trading time is (again bar weekends and holidays) equal to physical time. However, closure-closure studies show that the variance between successive days during the week can be up to 20% lower than similar values across weekends (Mantegna and Stanley, 2000). Market activity is often implicitly assumed to be uniform during trading hours but data shows that both the trading volume and number of contracts made vary. With tick data, it is possible to define time in terms of transactions. However, the volume of transactions remains as a random variable. In general, care should thus always be taken when comparing different results published in the literature.

2.1.2 Characterizing price fluctuations

Let us now have a closer look at various ways of dealing with time fluctuations of financial data. Suppose $s(t)$ is the price of an asset at time t . The simplest way to evaluate price fluctuations is to consider the change of the price over a time δt :

$$(2.1) \quad \delta s(t, \delta t) = s(t + \delta t) - s(t)$$

This has the merit of being simple and the price changes are additive,

$$(2.2) \quad \delta s(t, \delta t_1 + \delta t_2) = \delta s(t + \delta t_1, \delta t_2) + \delta s(t, \delta t_1).$$

One difficulty with this approach is, as we implied above, that the fluctuations are affected by changes in prices due to economic changes that occur during periods of inflation or recession. An alternative approach might be to incorporate some kind of inflation factor, $f(t)$, and use the renormalized price changes $\delta \tilde{s}(t, \delta t)$, defined as

$$(2.3) \quad \delta \tilde{s}(t, \delta t) = f(t + \delta t)s(t + \delta t) - f(t)s(t).$$

Of course the problem with this approach is that there is no obvious choice for the function $f(t)$. Another approach that is intuitively more reasonable is to compute the

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return, $r_0(t, \delta t)$,

$$r_0(t, \delta t) = \frac{\delta s(t, \delta t)}{s(t)} = \frac{s(t + \delta t)}{s(t)} - 1. \quad (2.4)$$

This gives the *relative* increase (profit) or decrease (loss) during the period, δt . The disadvantage with this approach is that returns defined in this way are not additive. They are multiplicatively linked, viz

$$[r_0(t, \delta t_1 + \delta t_2) + 1] = [r_0(t + \delta t_1, \delta t_2) + 1][r_0(t, \delta t_1) + 1]. \quad (2.5)$$

This problem can be resolved by using the logarithm of the price and defining the *log-price return*, $r(t, \delta t)$, as

$$r(t, \delta t) = \ln s(t + \delta t) - \ln s(t) = \ln \frac{s(t + \delta t)}{s(t)}. \quad (2.6)$$

The log-price returns are now seen to be additive,

$$r(t, \delta t_1 + \delta t_2) = r(t + \delta t_1, \delta t_2) + r(t, \delta t_1). \quad (2.7)$$

The incorporation of an inflation factor, $f(t)$, leads to

$$\ln \frac{f(t + \delta t)s(t + \delta t)}{f(t)s(t)} = r(t, \delta t) + \ln \frac{f(t + \delta t)}{f(t)}. \quad (2.8)$$

We see that for $f(t) = f(t + \delta t)$ the second term on the RHS is zero; the use of log-price returns takes account of inflation that remains constant over time.

Furthermore, if δt is small and the associated price change is also small, $|\delta s(t, \delta t)| \ll s(t)$, then we see from Eqns (2.4) and (2.6) that $r(t, \delta t) \sim r_0(t, \delta t)$.

A common measure of fluctuations is the absolute value of the log-price returns, $|r(t, \delta t)|$. This is closely related to the *instantaneous volatility*, defined as

$$\sigma_r(t, \delta t) = \sqrt{r(t, \delta t)^2 - \langle r(t, \delta t) \rangle^2}, \quad (2.9)$$

where $\langle r(t, \delta t) \rangle$ is the time average of the log price return,

$$\langle r(t, \delta t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T r(t, \delta t) dt. \quad (2.10)$$

In the case $\langle r(t, \delta t) \rangle = 0$ we thus obtain $\sigma_r(t, \delta t) = +\sqrt{r(t, \delta t)^2} = |r(t, \delta t)|$.

Figure 2.1 shows the variation of price $s(t)$, log-price return $r(t, 1 \text{ day})$, and absolute value of log-price returns $|r(t, 1 \text{ day})|$ for daily Dow Jones Industrial Average (DJIA) data over the period 2000 to mid-2011. An important feature of financial data, and clearly visible in Figure 2.1(c), is the occurrence of clustered regions where the volatility is much greater than the average value. We will return to this so-called *stylized fact* of financial data on various occasions throughout the book. It is not captured in the simple Brownian motion model of stocks, as discussed in Chapter 6.

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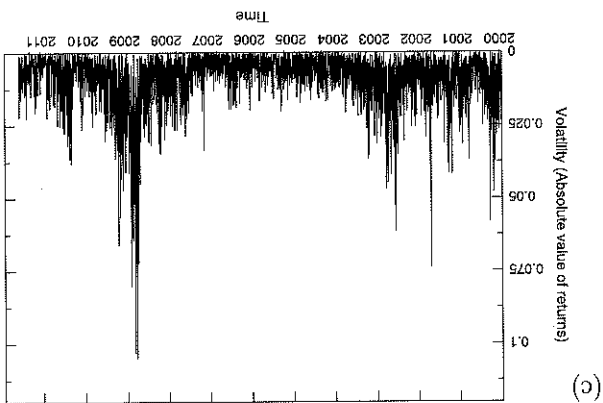
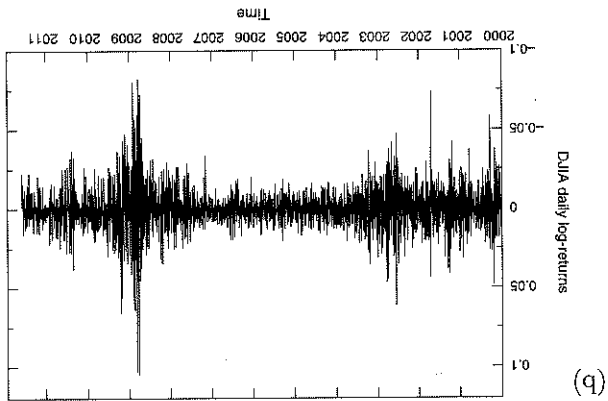
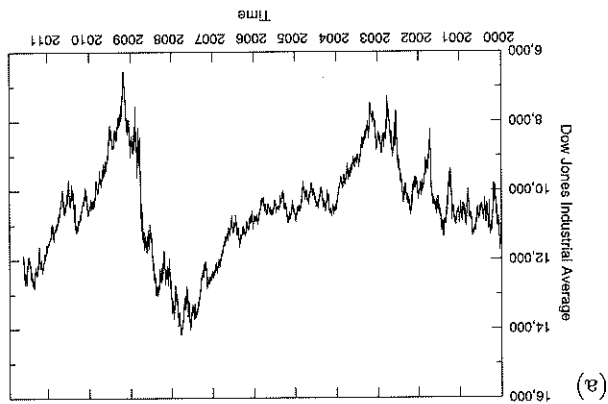


Figure 2.1 Daily data for the Dow Jones Industrial Average (DJIA) over the period 2000 to mid-2011: (a) price $s(t)$; (b) log-price return $r(t, 1 \text{ day})$; (c) absolute value of log-price returns $|r(t, 1 \text{ day})|$ (instantaneous volatility).

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2.1.3 Quantifying temporal correlations

A measure of how the value of a variable x at time t is influencing its value at time $t + \tau$ is given by the evaluation of the *autocorrelation function*, $G(\tau)$, defined as

$$G(\tau) = \langle x(t)x(t+\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x(t)x(t+\tau)dt. \quad (2.11)$$

In the case of a set of N discrete values, such as prices or returns recorded at times t_i , the RHS is approximated by $\frac{1}{N} \sum_i x(t_i)x(t_i + \tau)$. (Applied to values of log-price return we may write $G(\tau, \delta t) = \langle r(t, \delta t)r(t + \tau, \delta t) \rangle$.)

Provided the standard deviation, $\sigma_t = \sqrt{\langle x^2(t) \rangle - \langle x(t) \rangle^2}$ is non-zero and finite, it is also common to define a *normalized autocorrelation function* $R(\tau)$ by

$$R(\tau) = \langle (x(t) - \langle x(t) \rangle)(x(t+\tau) - \langle x(t+\tau) \rangle) \rangle / (\sigma_t \sigma_{t+\tau}) = (G(\tau) - \langle x \rangle^2) / (\sigma_t \sigma_{t+\tau}). \quad (2.12)$$

The values of R are in the interval $[-1:1]$, with 1 indicating perfect correlation, -1 perfect anti-correlation, and 0 no correlation.

The *autocovariance function* $C_{XX}(\tau)$ is defined by

$$C_{XX}(\tau) = \langle (x(t) - \langle x(t) \rangle)(x(t+\tau) - \langle x(t+\tau) \rangle) \rangle = \langle x(t)x(t+\tau) \rangle - \langle x(t) \rangle \langle x(t+\tau) \rangle, \quad (2.13)$$

i.e. $C_{XX}(\tau) = R(\tau)\sigma_t\sigma_{t+\tau}$.

For a stationary process, i.e. $\langle x(t) \rangle = \langle x(t+\tau) \rangle$ and $\sigma_t = \sigma_{t+\tau} = \sigma$, this reduces to

$$C_{XX}(\tau) = G(\tau) - \langle x(t) \rangle^2 = \sigma^2 R(\tau). \quad (2.14)$$

In order to evaluate correlations between *different* time series, e.g. correlations between different stocks, we will in Section 3.1.4 introduce a covariance matrix (see also Section 11.3).

For so-called *ergodic systems* a time average is equal to an ensemble average, see Section 1.4. We have in Eqns (2.11) and (2.14) already implicitly made this assumption when interchanging the definition of a temporal average (i.e. an integral over time) with the notation $\langle \dots \rangle$ which generally refers to an ensemble average.

In Figure 2.2 we show the (normalized) autocorrelation function $R(\tau, \delta t)$, eqn (2.12), for the log returns $r(t, \delta t)$ of the daily Dow Jones Industrial Average data of Figure 2.1. The function decays to zero essentially at the first time step and the daily returns appear to be independent. This is very similar to the case of having returns that are Gaussian distributed, see Figure 6.7, as is the case in the simple Brownian motion model of stocks which we will describe in Section 6.1.

However, the behaviour of the normalized autocorrelation function for the *volatility* is quite different. As a result of the clustered volatility that is evident in Figure 2.1(c), this function, shown in Figure 2.3, exhibits a very slow decay over quite a long time period of the order of a few hundred days. Clearly, the volatility fluctuations are *not* independent. Various research teams have suggested a power law decay with exponents between 0.2 and 0.3 over the first 100 days (Dacorogna *et al.*, 1993, Liu *et al.*, 1999, Roman *et al.* 2008, Shen and Zheng, 2009).

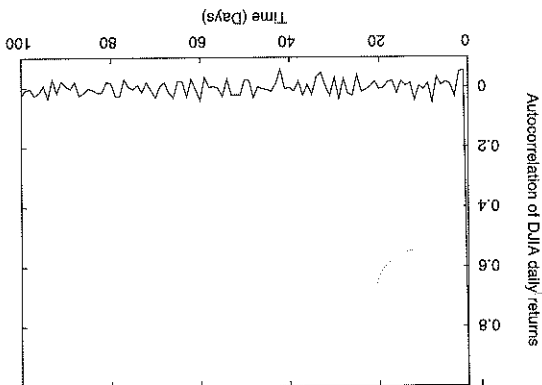


Figure 2.2 Plot of the normalized autocorrelation function $R(t, 1 \text{ day})$ of daily returns $r(t, 1 \text{ day})$ using Dow Jones Industrial Average data from 1993 to 2012. It is clear that there are no correlations certainly after the second day.

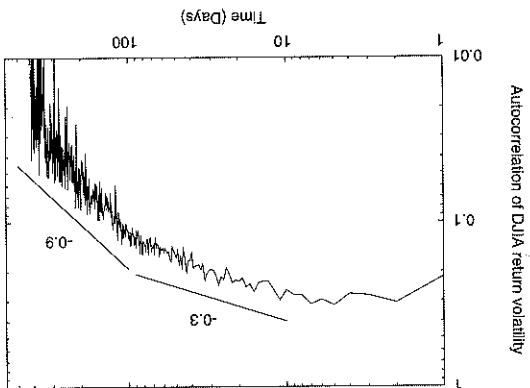


Figure 2.3 Auto-correlation $R(t, 1 \text{ day})$ of the *magnitude* (volatility) of daily DJIA log returns $|r(t, 1 \text{ day})|$ (data from 1993–2012), plotted on a log-log scale. Note that unlike the autocorrelation of the returns, here we see a very slow decay. Power law regimes may be identified for the ranges of 10–100 days, and exceeding 100 days, with approximate exponents of -0.3 and -0.9 (shown as lines of slope -0.3 and -0.9, respectively, in this double-logarithmic plot). We will return to a detailed analysis of this data in Sections 10.4 and 10.5.

We will in the following chapters detail the ‘econophysics approach’ of understanding the characteristic features of financial data, such as clustered volatility and slowly decaying autocorrelation for volatility. However, first we shall briefly describe some ways in which investors might assess financial data.

in Figure 6.3(b).

this data we compute 994 log-returns $r(t, 1\text{month}) = \ln(s_{t+1}/s_t)$, which are displayed in Figure 6.3(a). From 1928 to August 2011, i.e. a total of 995 data points, as shown in Figure 6.3(a). From the financial data that we have chosen is monthly DJIA data for the period October 1928 to August 2011, i.e. a total of 995 data points, as shown in Figure 6.3(a). From

At this stage it is useful to show some numerical simulations based on the model developed in this chapter and see how well it is suited to describe actual financial data. To this purpose we have written C program code, and analysed both simulation and financial data using the free graphics software Xmgrace², which features a number of statistical tools. (Software packages such as Mathematica or Excel are equally suitable.)

6.3 Comparing computer simulations for the geometric Brownian model with real stock data

6.3 Comparing computer simulations for the geometric Brownian

The books and papers concerned with theoretical finance and stochastic processes derive these results via rather formal developments in stochastic calculus originally due to Ito. We return to the topic in Chapter 7, where we derive the results of Ito without recourse to stochastic methods. We further shall show that the result of Ito is in fact limited to the special case of Gaussian fluctuations and obtain a more generalized expression valid for more generalized stochastic processes.

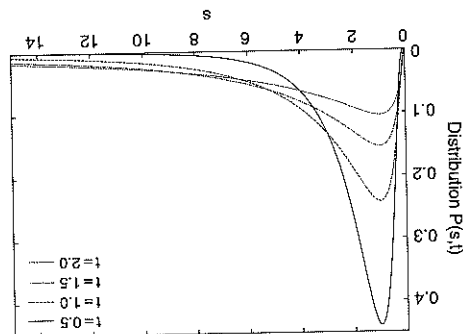
So we see that whilst average log-prices increase linearly with time at a rate, v_d , prices themselves increase *exponentially* at a rate $v_{d,s} = v_d + \sigma_0^2/2$. This is important from an empirical point of view when drawing conclusions from various data sets plotted as log price and/or price versus time.

$$\langle s(t)^2 \rangle - \langle s(t) \rangle^2 = \exp[2v_{d,s}t](\exp[\sigma_0^2 t] - 1). \quad (6.21)$$

Exercise 6.3 Performing a similar computation as above, the reader should show that the variance of prices is given by

(here we set $\sigma_0^2 = 1, v_d = 1$).

Figure 6.2 The log-normal distribution $P(s, t)$ of eqn (6.17), plotted for several values of t



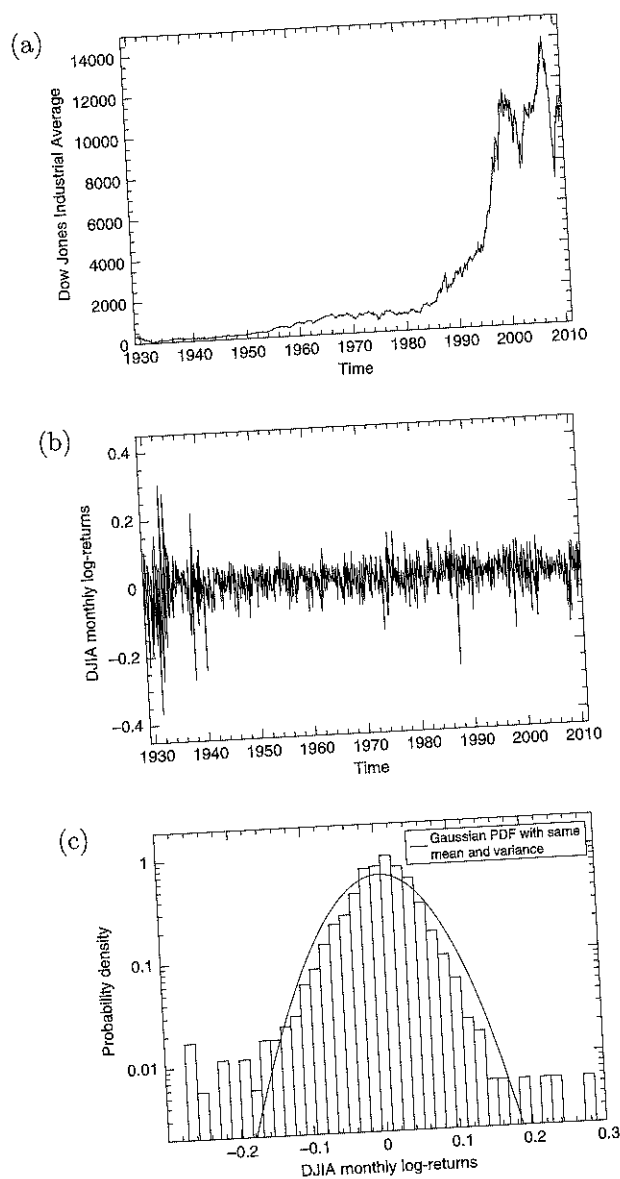


Figure 6.3 The Dow Jones Industrial Average over the period October 1928 to August 2011. (a) Shown is monthly data, i.e. a total of 995 data points, corresponding to the closing price of the first day of every month. (b) The monthly variation of the log-return shows periods of high volatility. (c) The probability distribution of these log-returns is only inadequately described by a Gaussian, which fails to capture the large fluctuations seen in the data.

An empirical probability distribution of these values is obtained by counting the number of log-returns that lie in a certain interval $[r, r + dr]$. The result of this binning procedure is shown in Figure 6.3(c), where we have chosen a logarithmic scale for the ordinate. In such a semi-log plot, a Gaussian distribution, (see eqn (3.10)) turns into an inverted parabola, as can be seen by taking the logarithm, i.e.

$$\ln p(r) = -\frac{\ln(2\pi\sigma^2)}{2} - (r - m)^2/(2\sigma^2). \quad (6.22)$$

The mean m and variance σ^2 of the log-returns can be directly computed from our data in Figure 6.3(b), resulting in $m = 0.0039$ and $\sigma^2 = 0.0542$. The solid line in Figure 6.3(c) shows a Gaussian distribution, eqn (3.10), with these parameters. Referring to eqn (6.11) this corresponds to $m = v\delta t$ and $\sigma^2 = \sigma_0^2\delta t$ where $\delta t = 1$ (month). This is clearly a poor description for log-returns outside the interval of say $[-0.15; 0.15]$. The probability of having returns of magnitude larger than 0.15 (as feature in Figure 6.3(b)) is drastically underestimated. A closer look at Figure 6.3(b)) also shows that fluctuations are often clustered; we will return to this observation shortly. Let us now turn to computer simulations of the Brownian motion model. Using a random number generator we create 994 Gaussian distributed random numbers with mean $m = 0.0039$ and variance $\sigma^2 = 0.0542$, i.e. the values obtained from the DJIA

data. We choose these random numbers to be our log-returns $r(t, 1) = \ln(s_{t+1}/s_t)$ and display them in Figure 6.4(a), where 'time' is simply the sequence of their generation. We can see that very few log-returns are greater than say $3\sigma = 0.16$; large fluctuations relative to the mean value do not occur too often in Gaussian statistics.

The empirical probability distribution of these random numbers is shown in Figure 6.4(b), again with a logarithmic scale for the ordinate. As expected by the construction of our random numbers, they are normally distributed and well described by eqn. (6.22). Having generated Gaussian distributed log-returns we can finally use these to construct the time series of the price of an imaginary Gaussian asset or an index by simply summing up the computed increments $\ln(s(t) - 1)/s(t)$ (starting with the log of the October 1928 closing price of the DJIA) and taking the exponential of the value at every time-step. The result is shown in Figure 6.4 for our total of 994 increments, corresponding to 994 unit time steps.

Without detailed analysis, it is easy to imagine that the simulated graphs of Figure 6.4(c) might somehow correspond to the real data displayed in Figure 6.3(a). However, it needs to be remembered that Figure 6.4(c) shows only *one realisation* of a random process. Performing the simulation again with different random numbers, although chosen from the same distribution, will result in a different realisation, such as the one shown in Figure 6.5. This could result in better or less visual similarity to the real DJIA data.

A more important issue concerning the value of our simulation has to do with the character of the fluctuations of the log-returns. As we already mentioned earlier, our random data does not reproduce the clustering observed in the DJIA data. This becomes more obvious when comparing graphs showing the modulus or the square of the log-return as a function of time, as in Figure 6.6. (Note that the clustering is more pronounced in our daily DJIA data, shown in Figure 2.1.)

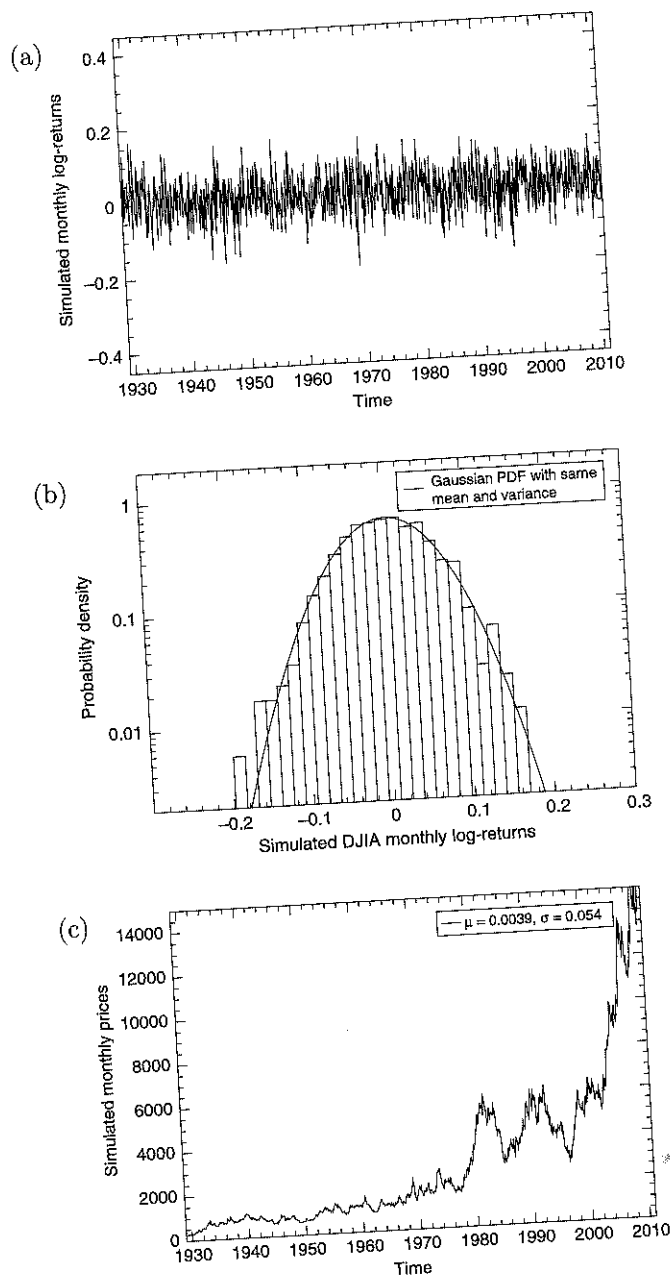


Figure 6.4 Computer simulation using the Brownian motion model of stock prices. (a) Gaussian distributed log-returns with values of mean $m = 0.0039$ and variance $\sigma^2 = 0.054^2$, as obtained from DJIA data from October 1928 to August 2011 (see Figure 6.3). There are hardly any log-returns outside the interval $[-0.15; 0.15]$. (b) Histogram of the simulated log-returns, together with the Gaussian distribution from which they have been chosen. (c) Simulation of a stock index as obtained from consecutively adding up the log-returns of (a) and taking exponentials. A certain visual similarity with the DJIA data of Figure 6.3(a) may not detract from the dissimilarities with respect to the shape of the distribution function of the log-returns and the lack of clustering in this simulation data.

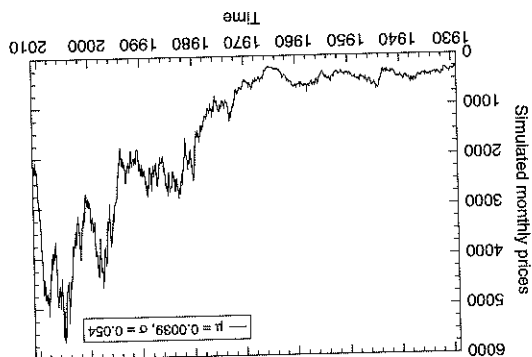


Figure 6.5 Example of a further computer simulation using the same Gaussian distribution as used for the simulation in Figure 6.4 but a different set of random numbers. The visual similarity with the real data shown in Figure 6.4 is less obvious in this case.

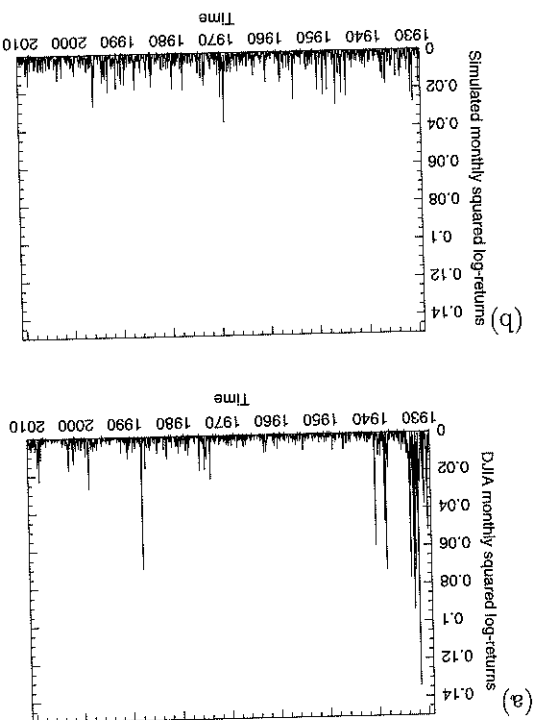


Figure 6.6 The squared values of log-return (a measure of volatility) for both the DJIA monthly data of Figure 6.3(b) and the simulated data of Figure 6.4(a). The volatility of the financial data shows clustering which is absent in the simulated data. This clustering (which is much more pronounced in daily data, see Figure 2.1) is responsible for the slow decay in the autocorrelation function for the squared values of return, $H_2^2(\tau)$ as shown in Figure 6.8(a).

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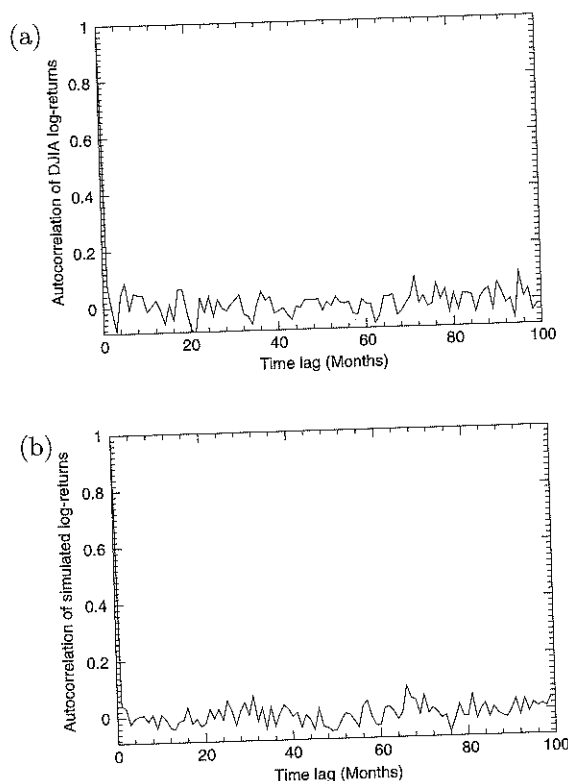


Figure 6.7 Variation of the normalized autocorrelation function $R_r(\tau)$ of log-returns. (a) It is known from minute-by-minute empirical data that $R_r(\tau)$ decays over a time span of approximately 30 minutes, so there are no correlations in the shown *monthly* DJIA data. (b) As expected there are no correlations in the data produced from a computer simulation of the simple Brownian model.

In order to have a closer look at time correlations we will now consider both autocorrelation function of log-returns and the square of log-returns, a measure of volatility. Figure 6.7 shows the normalised autocorrelation function $R_r(\tau)$ (see eqn (2.12)) for the log-price return for both monthly DJIA log-returns and for our simulated log-returns. In both cases R_r fluctuates about zero already after one month (i.e. one time step in the simulation). It is important to note though, that autocorrelations do exist in financial returns. However, they are on time scales of the order of minutes and are thus not visible in the monthly data shown here. We will return to this in Chapter 10 where we discuss minute by minute Dow Jones data.

The situation is different when comparing the normalised autocorrelation function for *squared* returns, $R_{r^2}(\tau)$, as shown in Figure 6.8. In this case the DJIA data shows positive autocorrelation for up to about 40 months, reflecting the clustering in the volatility as we remarked earlier. As expected from the way it has been constructed,

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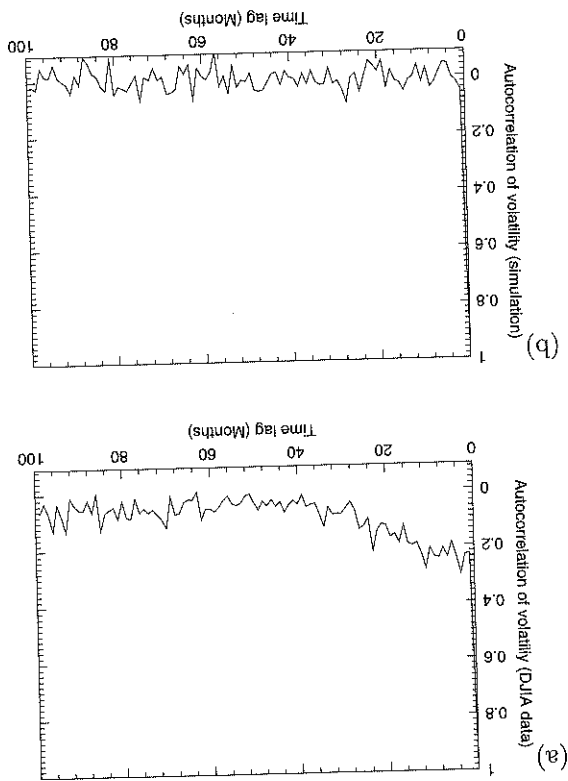


Figure 6.8 Variation of the normalized autocorrelation function $R_{\nu^2}(\tau)$ of the volatility. (a) The log-returns for monthly DJIA data show positive correlations for up to about 40 months with a sharp drop over the first two months followed by a roughly linear decay. (b) Correlations are absent in the data for log-returns produced using the simple Brownian motion model.

there are no correlations in our simulated data. We had commented on the slowly decaying autocorrelation of the volatility already in Section 2.1.3 and noted that this decay may feature regimes approximated by power laws (see Figure 2.3). The time resolution of our monthly data shown in Figure 6.8(a) is not sufficient to resolve this. We will in Chapter 10 return to this issue when examining minute-by-minute data.

6.4 Issues arising

The comparison of monthly DJIA data with data produced from the simple Brownian motion model of log-returns has shown that while the model serves as a good 'first approximation' of the data, it has severe shortcomings.

- A Gaussian distribution does not describe the observed large values of log-returns that are found in empirical (monthly) data. (*Annual* log-returns are reasonably well described by a Gaussian, see Figure 6.1.)

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